

DEVIKA NAIR

devikn@uw.edu | [LinkedIn](#) | [Google Scholar](#) | [GitHub](#)

M.S. ECE student at the University of Washington (GPA 3.77) with 3+ years of embedded firmware experience at SanDisk (NAND/SSD, eMMC/UFS). Strong in low-level C/C++, RTOS, FPGA/RTL, and hardware bring-up and debug. Seeking a Fall 2026 embedded / hardware internship.

EDUCATION

University of Washington <i>M.S., Electrical and Computer Engineering — GPA: 3.77/4.00</i> • Relevant coursework: Digital System Design–FPGA; Computer System Architecture; Embedded & Real-Time Systems; Deep Learning for Embedded/Real-Time Intelligence; VLSI Design.	Seattle, WA <i>Sept. 2025 – Jun. 2027 (Expected)</i>
MKSSS Cummins College of Engineering for Women <i>B.Tech., Electronics and Telecommunications — CGPA: 7.71/10</i> • Relevant coursework: VLSI Design; Microcontrollers & Embedded Systems; Digital Signal Processing; Computer Networks; Data Structures & Algorithms; Control Systems.	Pune, India <i>Jul. 2017 – Jun. 2021</i>

EXPERIENCE

SSD Firmware Engineer Intern <i>SanDisk</i> • Developing firmware telemetry and physical storage algorithms for enterprise SSD products. • Applying prior NAND flash, reliability, and firmware experience (eMMC/UFS, CVD analytics) to enterprise SSD (NVMe/PCIe) development.	Milpitas, CA <i>Jun 2026 – Sep 2026</i>
Research Assistant — UW SEAL Lab (Prof. Alex Mamishev) <i>University of Washington</i> • Built a real-time embedded sensing system for gas-leak detection on ESP32-S3 (Xtensa LX7) in C/RTOS: interrupt-driven, DMA-based acquisition over SPI/I ² C for kHz-rate MEMS data, with a low-latency DSP pipeline (FFT, RMS, feature extraction) and end-to-end firmware through WiFi/LoRa transmission.	Seattle, WA <i>Winter 2026 – Spring 2026</i>
Firmware Engineer → Senior Firmware Engineer <i>SanDisk (Western Digital)</i> • Drove end-to-end NAND flash (eMMC/UFS) feature implementations for a major automotive client, enabling on-time launch (\$45M revenue, zero customer-reported issues); improved NAND endurance by 25% and reduced latency variation by 15%. • Built a Cell Voltage Distribution (CVD) analytics framework integrating read-retry and ECC telemetry, enabling voltage-drift modeling and accelerating root-cause debug by ≈40%. • Contributed to ASPICE CL3 certification (an industry first in embedded storage) and published a U.S. Patent on predictable low-latency NAND access; delivered a firmware MVP unlocking \$10M in revenue. • Supported bring-up, validation, and MISRA-C compliance for automotive-grade NAND reliability features.	Bengaluru, India <i>Jul 2021 – Jul 2025</i>

TECHNICAL SKILLS

Languages: C, C++, Python, Verilog, Bash/Shell, MATLAB, SQL
Embedded & Firmware: ARM Cortex-M/A; bare-metal/low-level C; RTOS (FreeRTOS, ThreadX); interrupts, DMA; Linux drivers, U-Boot, linker scripts; NAND/SSD eMMC/UFS; reliability (Refresh Manager, EPWR, CVD); MISRA-C/Parasoft; bring-up, validation, unit testing
Hardware / FPGA / EDA: Verilog RTL, Vivado, ModelSim; VLSI fundamentals; JTAG/SWD; logic analyzers, oscilloscopes
Build & Tools: Git, GCC, GDB, Make, CMake, Jenkins, Jira, VS Code, Vim
Protocols: I²C, SPI, UART, eMMC, UFS, PCIe, NVMe, TCP/IP

PROJECTS

Context-Switch Latency Characterization on Raspberry Pi 4 <i>C, Linux, POSIX, ARM</i> • Benchmarked Linux scheduling on ARM Cortex-A72 (SCHED_OTHER vs. SCHED_FIFO) with POSIX semaphores in C11, measuring context-switch latency (5.61–10.55 μs) to bound soft real-time behavior.	2025
Heuristic Write-Amplification Analysis for Flash Storage <i>MATLAB, Python, NAND FTL Simulation</i> • Heuristic firmware model to reduce NAND write amplification via adaptive data placement and wear-leveling (no ML dependencies).	2025

LEADERSHIP

Corporate Relations Representative — IEEE Women in Engineering (WIE) <i>University of Washington — lead industry/sponsor outreach for the student board</i>	Seattle, WA <i>2025 – 2026</i>
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